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DESIGN AND ANALYSIS OF ALGORITHM

DIVIDE AND CONQUER METHOD

Problem Statement 1:
Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format:
First Line Contains Integer m – Size of array
Next m lines Contains m numbers – Elements of an array

Output Format:
First Line Contains Integer – Number of zeroes present in the given array.

ANSWER:

```
#include <stdio.h>
int zeros(int *arr, int m){
   int left = 0, right = m-1, mid;
   while(left <= right){
      mid = (left+right) / 2;
      if(arr[mid] == 1) left = mid+1;
      else right = mid-1;
   }
   return m-left;
}
int main(){
```

```
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++) scanf("%d", &arr[i]);

    printf("%d", zeros(arr, n));
    return 0;
}
```

PROBLEM STATEMENT 2:

Given an array nums of size n, return the majority element.

The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.

Example 1:

Input: nums = [3,2,3]

Output: 3

Example 2:

Input: nums = [2,2,1,1,1,2,2]

Output: 2

Constraints:

- **n == nums.length**
- **$1 <= n <= 5 * 10^4$**
- **$-2^{31} <= nums[i] <= 2^{31} - 1$**

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For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

ANSWER:

```
#include <stdio.h>


int majority(int* arr, int n){
    int count = 0, first = arr[0], second;
    for(int i = 0; i < n; i++){
        if(arr[i] == first) count++;
        else second = arr[i];
    }
    if(count > n/2) return first;
    else return second;
}


int main(){
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++) scanf("%d", &arr[i]);
    printf("%d", majority(arr,n));
```

```
    return 0;
}
```

PROBLEM STATEMNT 3:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array
Next n lines Contains n numbers – Elements of an array
Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer:

```
#include <stdio.h>
int findFloor(int arr[], int low, int high, int x) {
    if (low > high) return -1;
    if (arr[low] > x) return -1;
    if (arr[high] <= x) return arr[high];

    int mid = low + (high - low) / 2;

    if (arr[mid] == x) return arr[mid];

    if (arr[mid] < x) {
        if (mid == high || arr[mid + 1] > x) return arr[mid];   // catch it here
```

```
        return findFloor(arr, mid + 1, high, x);
    } else {
        if (mid > low && arr[mid - 1] <= x) return arr[mid - 1];
        return findFloor(arr, low, mid - 1, x);
    }
}
int main(){
int n;
scanf("%d",&n);
int arr[n];
for(int i=0;i<n;i++)scanf("%d",&arr[i]);
int x;
scanf("%d",&x);
int result=findFloor(arr,0,n-1,x);
printf("%d\n",result);
return 0;
}
```

Problem Statement 4:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.
Note: Write a Divide and Conquer Solution

Input Format
First Line Contains Integer n – Size of array
Next n lines Contains n numbers – Elements of an array
Last Line Contains Integer x – Sum Value

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Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer:

```
#include <stdio.h>

int found = 0;

void findPair(int arr[], int low, int high, int x) {
    if (low >= high) {
        return;
    }

    int sum = arr[low] + arr[high];

    if (sum == x) {
        printf("%d\n", arr[low]);
        printf("%d\n", arr[high]);
        found = 1;
        return;
    } else if (sum < x) {
        findPair(arr, low + 1, high, x);
    } else {
        findPair(arr, low, high - 1, x);
    }
}
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    findPair(arr, 0, n - 1, x);

    if (!found) {
        printf("No\n");
    }

    return 0;
}
```

PROBLEM STATEMENT 5:

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:

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Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

ANSWER:

```
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b)
{
   return *(int *)a - *(int *)b;
}

int main(){
   int n;
   scanf("%d", &n);
   int arr[n];
   for(int i = 0; i < n; i++) scanf("%d", &arr[i]);

   qsort(arr, n, sizeof(int), compare);
```

```
    for(int i = 0; i < n; i++) printf("%d ", arr[i]);


    return 0;
}
```